what's under the switch?

digitalisierung und programmierung · prof. dr. nicolas meseth

1. what you don't need to know
2. your own stack
3. swap the bottom
4. what it costs

part 1

what you don't need to know

what happens between your finger and the light?

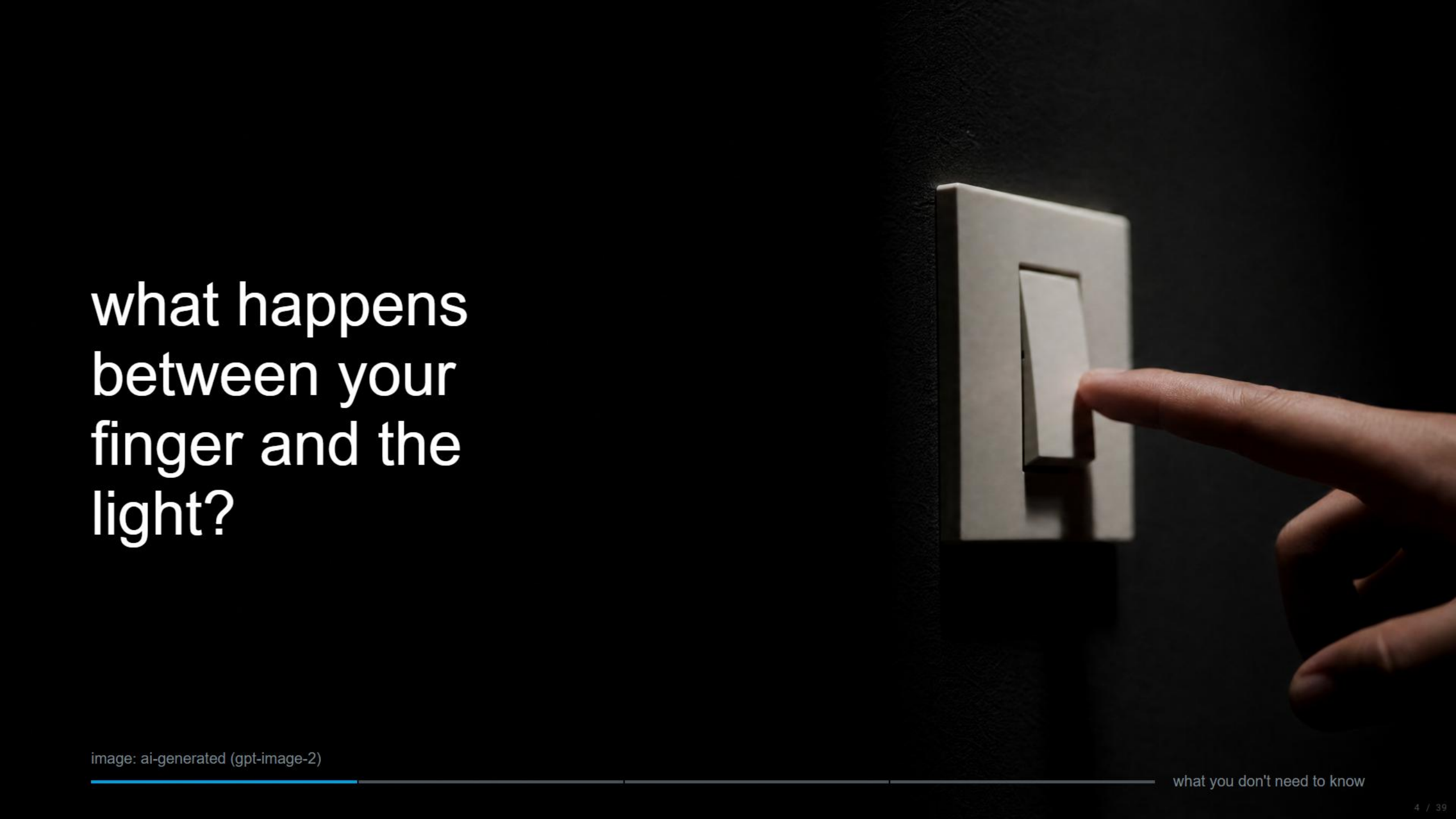
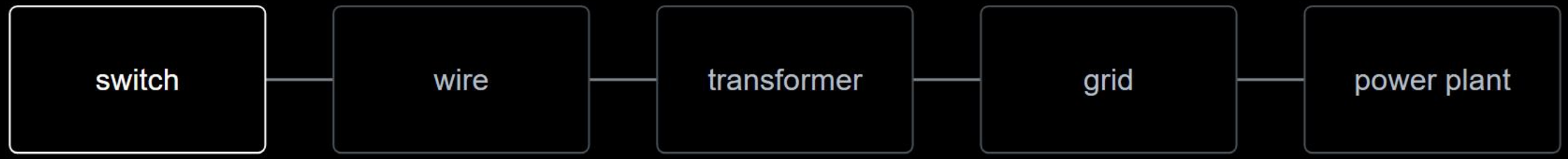


image: ai-generated (gpt-image-2)

behind the switch

behind the switch



you never think about any of this

the switch is the whole contract: up, down.

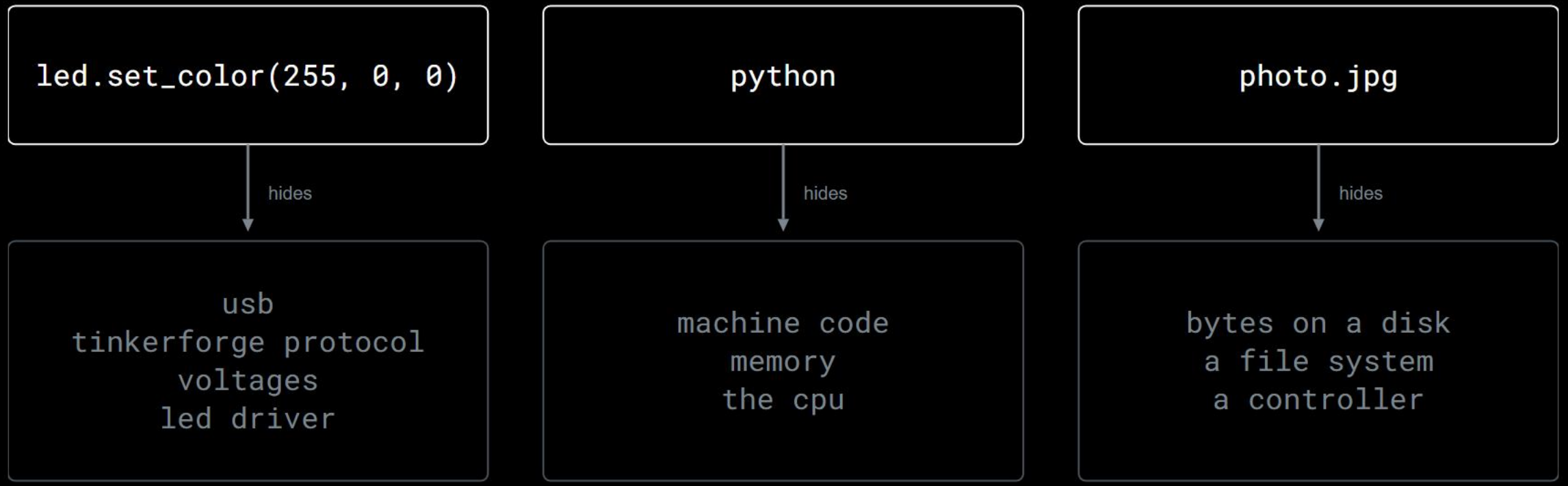
so here is the word for it

an abstraction hides how something works and shows what you can do with it.

the part it shows you is called the interface.

what you don't need to know

three you used today without noticing



part 2

your own stack

from a photo to light, one layer at a time

sending

meaning

your photo

your own stack

from a photo to light, one layer at a time

sending

meaning

code system

your photo

`FF D8 FF E0 …`

your own stack

from a photo to light, one layer at a time

sending

meaning

code system

frame

your photo

`FF D8 FF E0 …`

pre	type	len	payload	crc	end

your own stack

from a photo to light, one layer at a time

sending

meaning

code system

frame

bits

your photo

`FF D8 FF E0 …`

pre	type	len	payload	crc	end

`1101 0010 0110 1001`

from a photo to light, one layer at a time

sending

meaning

code system

frame

bits

symbols

your photo

FF D8 FF E0 …

pre | type | len | payload | crc | end

1101 0010 0110 1001

from a photo to light, one layer at a time

sending

meaning

code system

frame

bits

symbols

carrier

your photo

`FF D8 FF E0 …`

pre	type	len	payload	crc	end

`1101 0010 0110 1001`

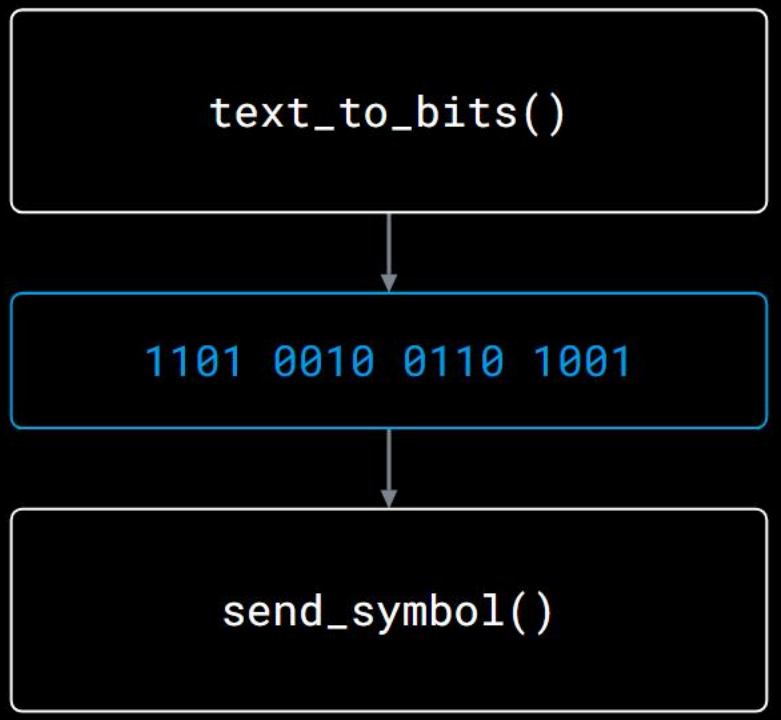
light

each layer talks only to the one directly below it.

that single rule is what makes everything else possible.

what that buys you

knows nothing about colours

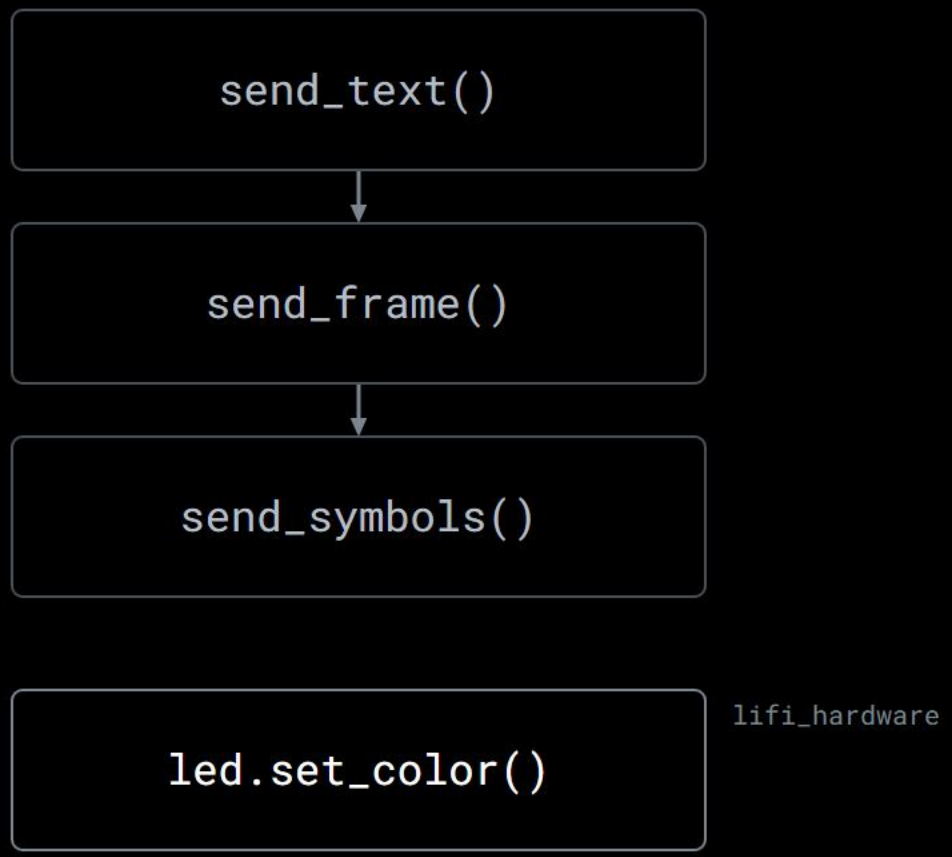


bits: all they share

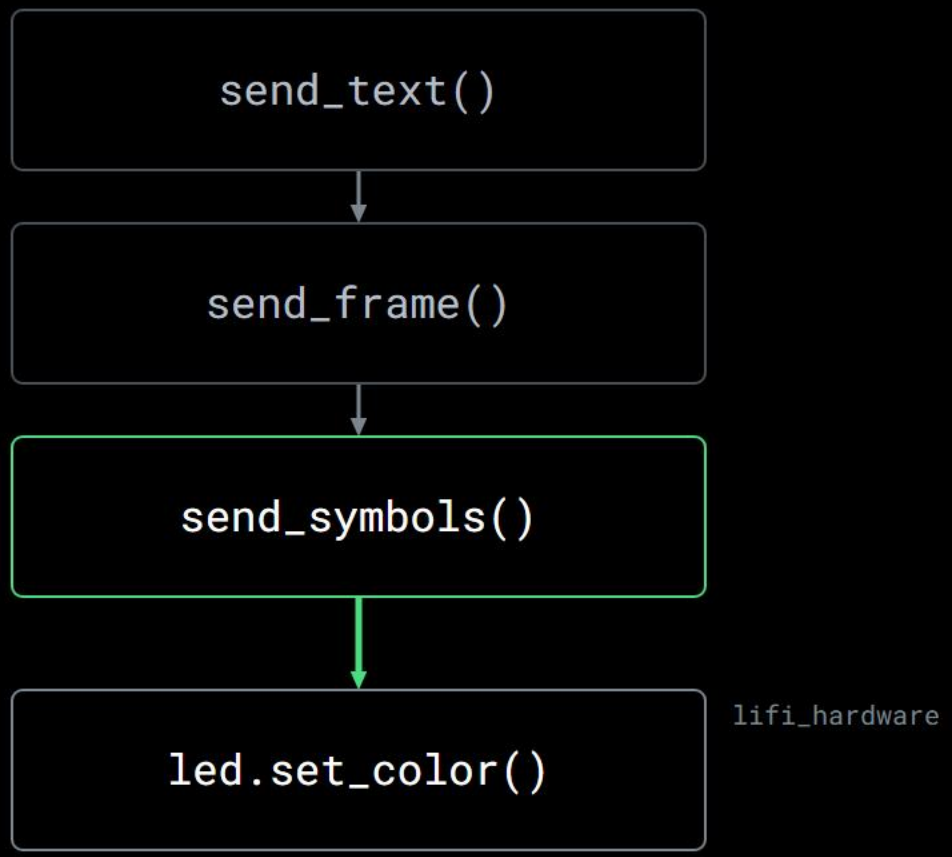
knows nothing about letters

not tidiness. this is why one can change without the other.

which of these may talk to the led?



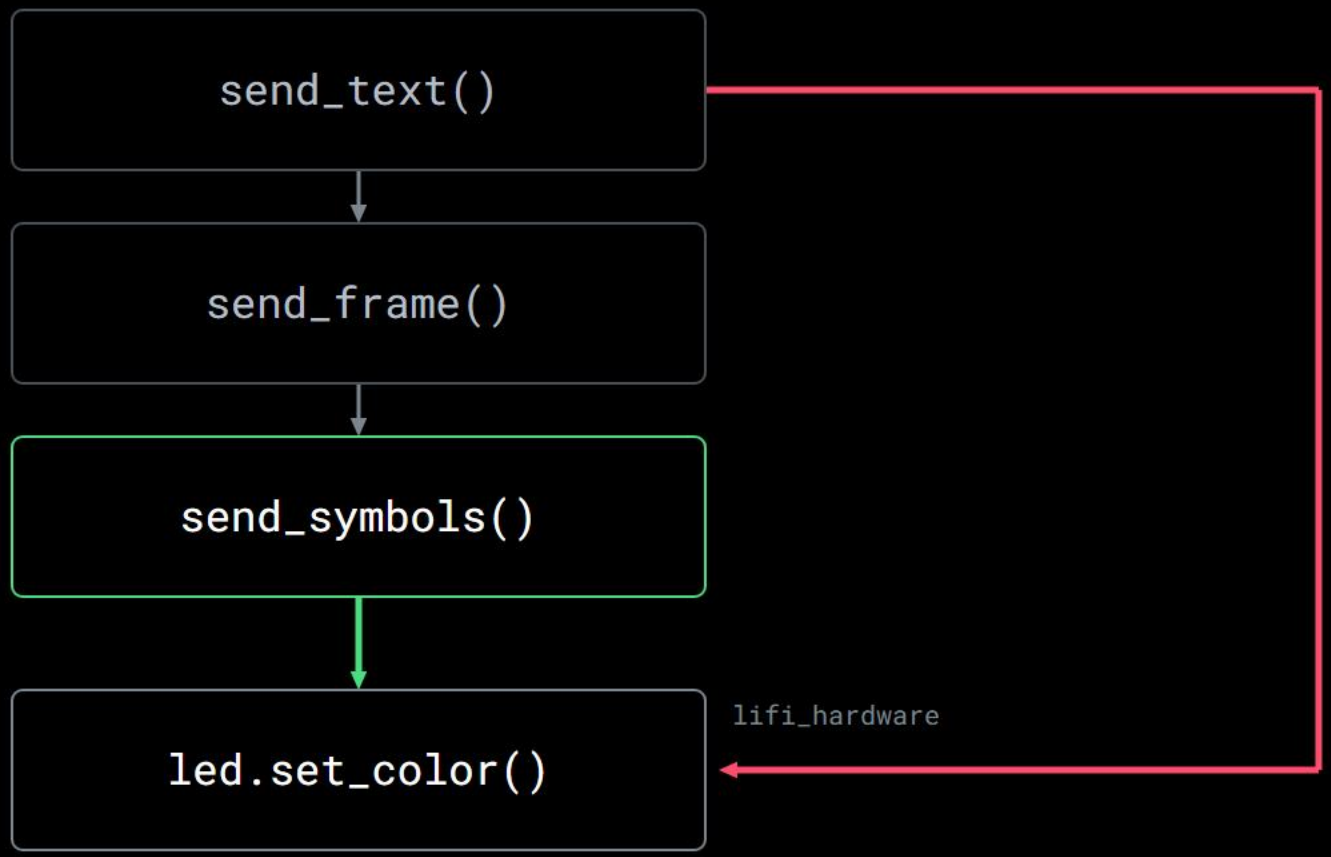
which of these may talk to the led?



allowed

only this layer knows which colour a symbol is

which of these may talk to the led?



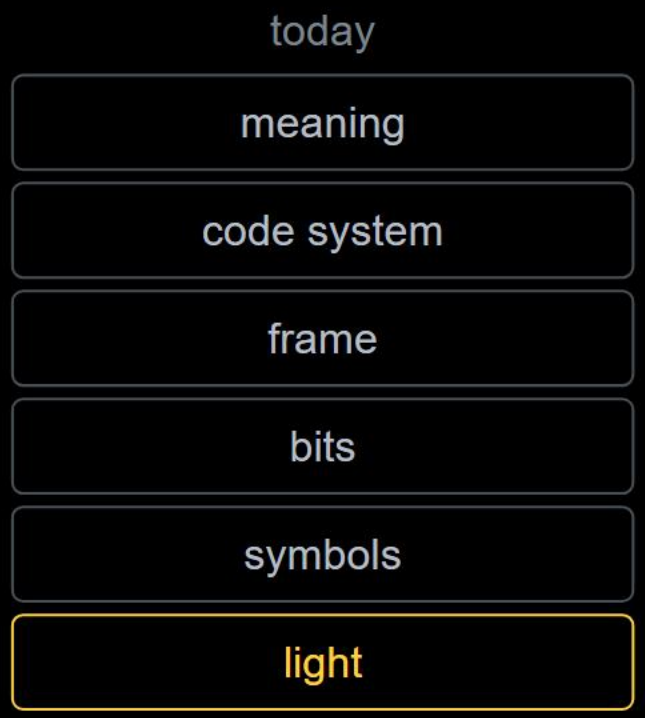
the shortcut
colours end up in
`send_text`

allowed
only this layer knows which colour a symbol is

part 3

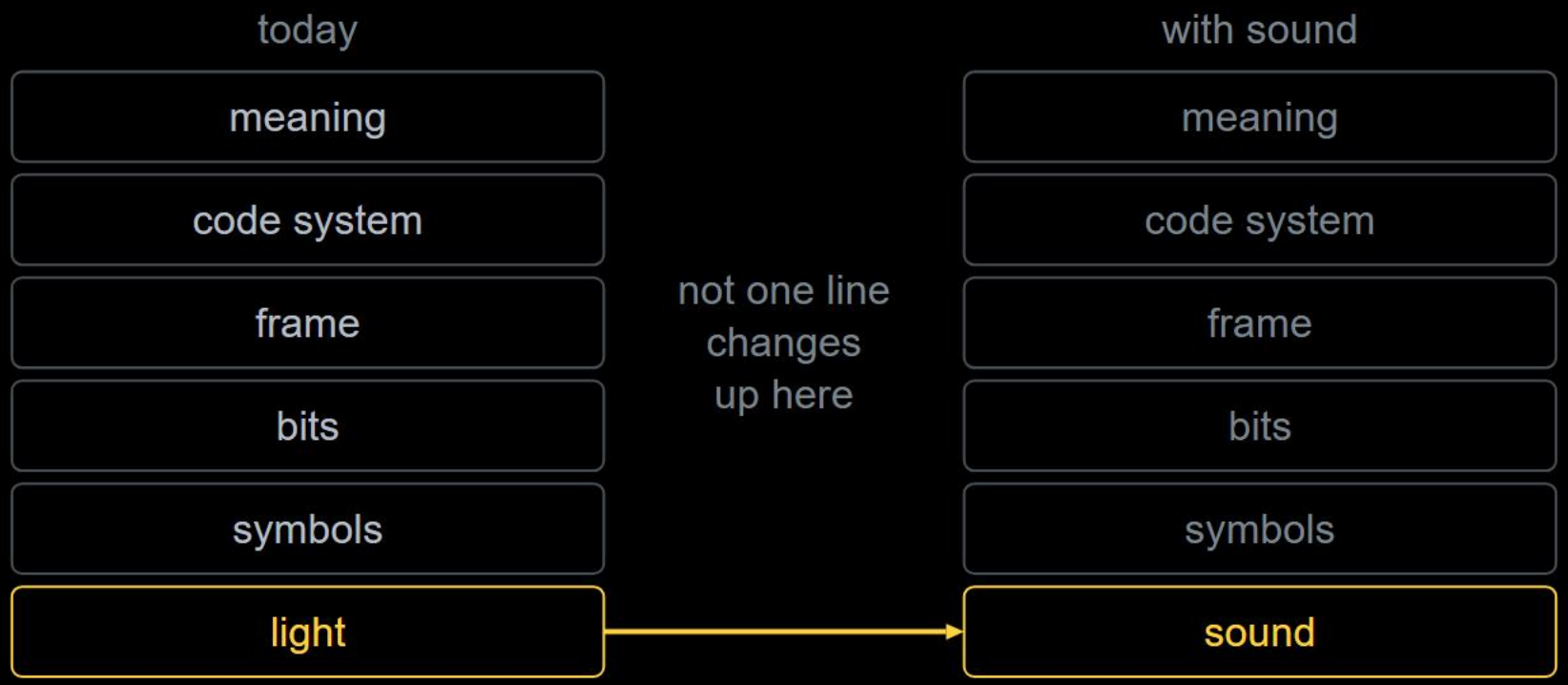
swap the bottom

same stack, different bottom



swap the bottom

same stack, different bottom



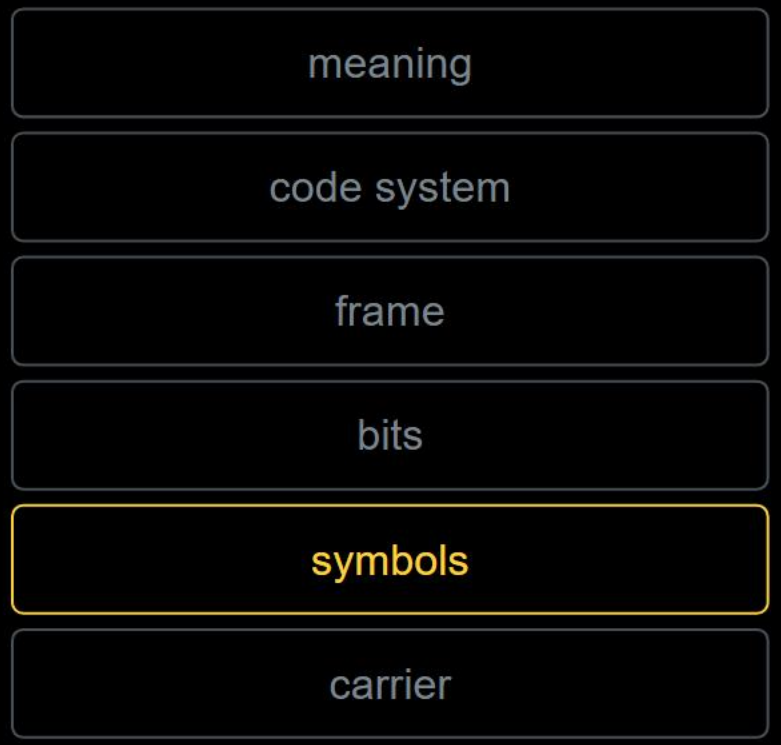
five layers unchanged. one exchanged.

you extend your alphabet from four colours to eight

- meaning
- code system
- frame
- bits
- symbols
- carrier

swap the bottom

you extend your alphabet from four colours to eight



colour table
calibration
bits ↔ symbols

every other layer: unchanged

a frame is defined in bits, not in colours

and it says nothing about time.

time belongs to the path, not to the message. that is why sound works.

everything here sits on light.

everything here sits on light.

light is the layer you could replace.

part 4

what it costs

1 **every layer adds its own data**

the frame costs bytes you cannot use

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2 **every boundary hides a knob**

the one you need is always the hidden one

1 **every layer adds its own data**

the frame costs bytes you cannot use

2 **every boundary hides a knob**

the one you need is always the hidden one

3 **you must learn where the boundaries are**

a wrong guess sends you looking in the wrong place

wrong letters arrive. where do you look first?

meaning · code system · frame · bits

symbols

carrier

?

wrong letters arrive. where do you look first?

meaning · code system · frame · bits

symbols ✓

carrier

tested: 100 % of symbols recognised correctly

wrong letters arrive. where do you look first?

meaning · code system · frame · bits	look here the fault is above the tested layer
symbols ✓	tested: 100 % of symbols recognised correctly
carrier	

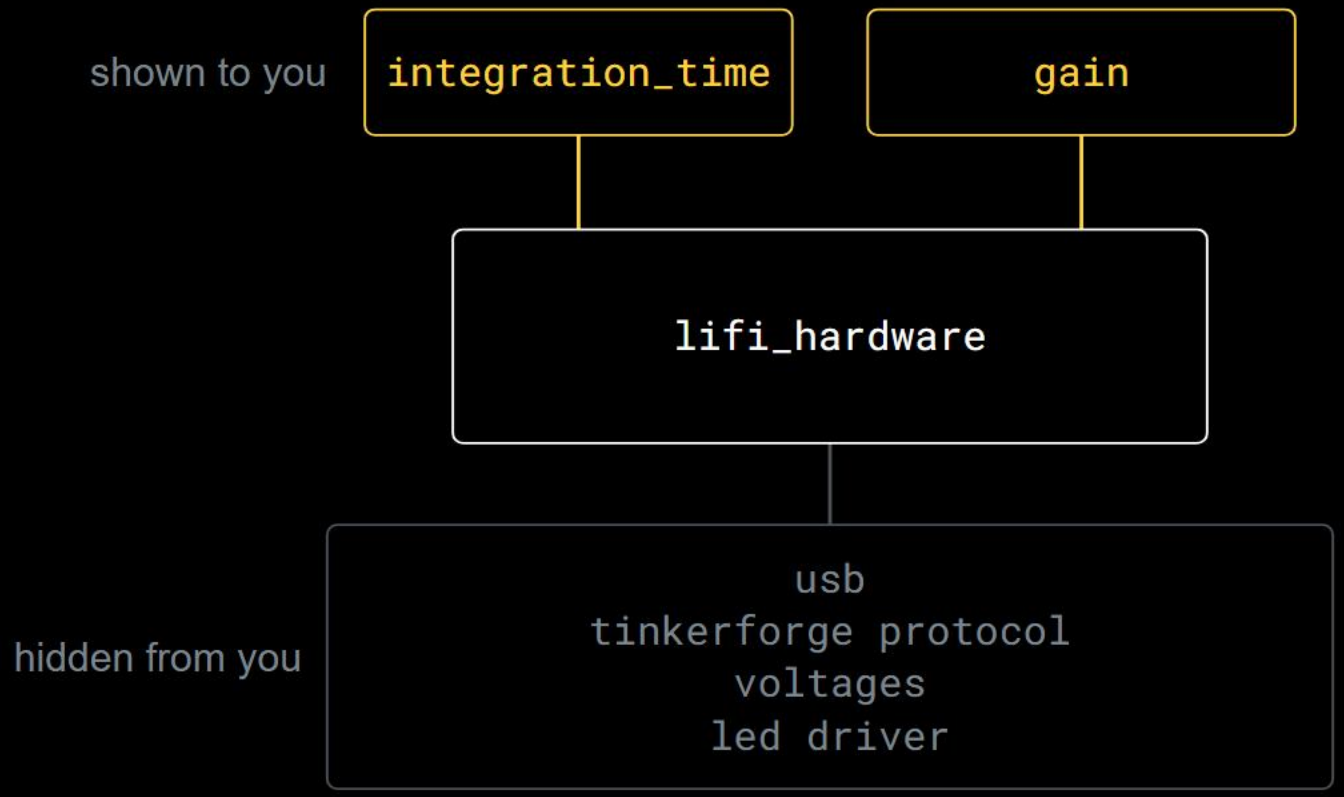
a tested layer is a layer you can stop suspecting.

how to test one layer

feed it a known input, check its output, with nothing below it.

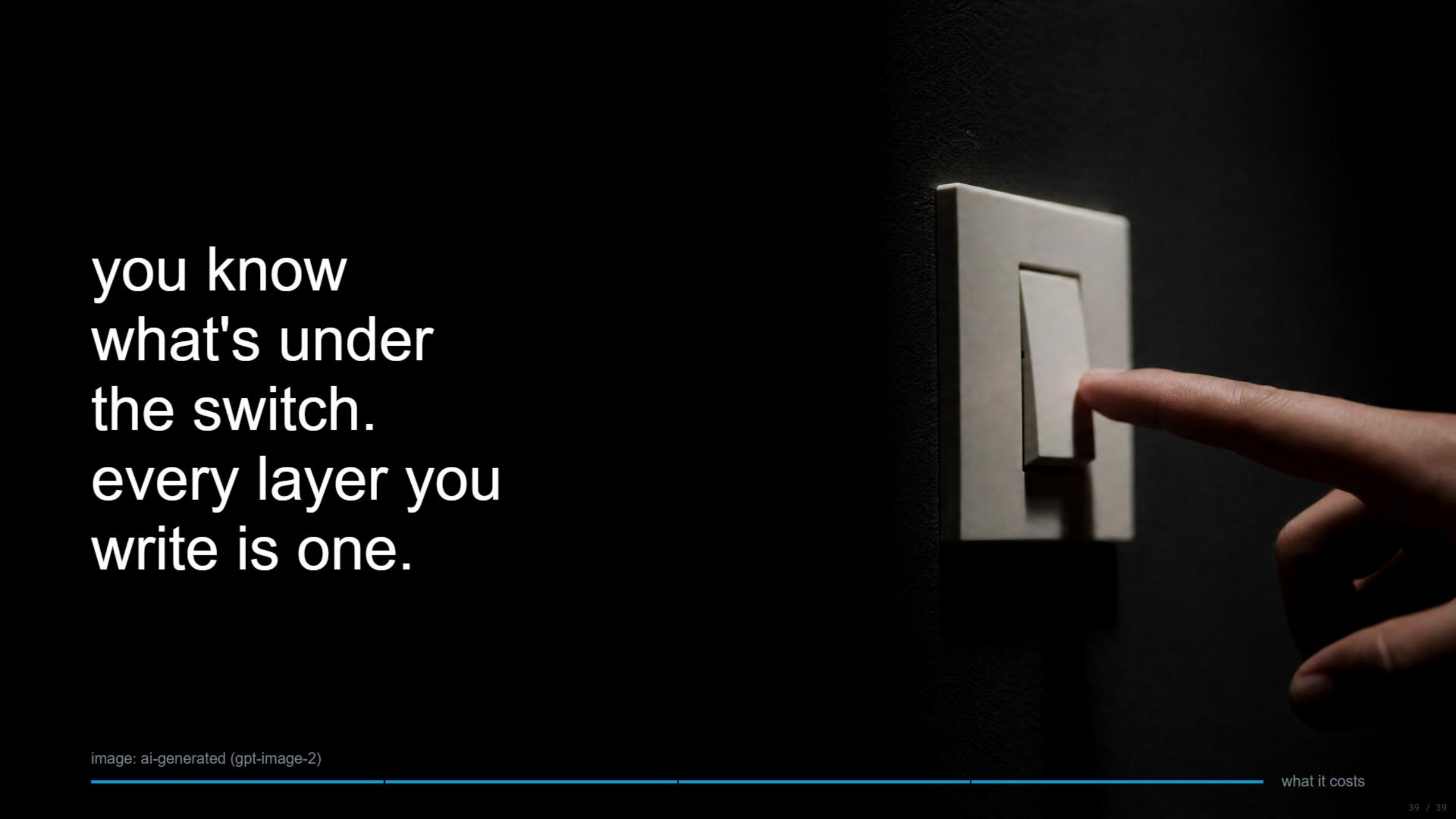
bits in, bits out. no led, no sensor, no room light.

why doesn't it hide everything?



a good interface hides what you don't need and shows what you must decide.

you know
what's under
the switch.
every layer you
write is one.



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